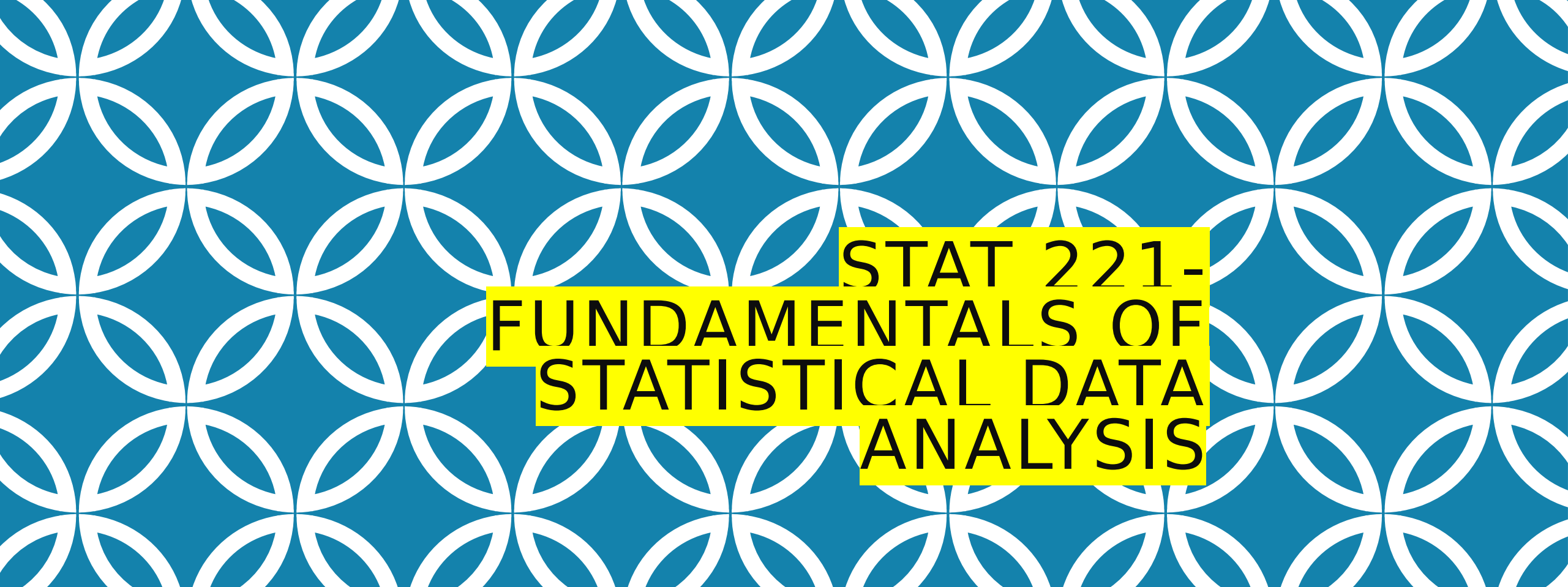
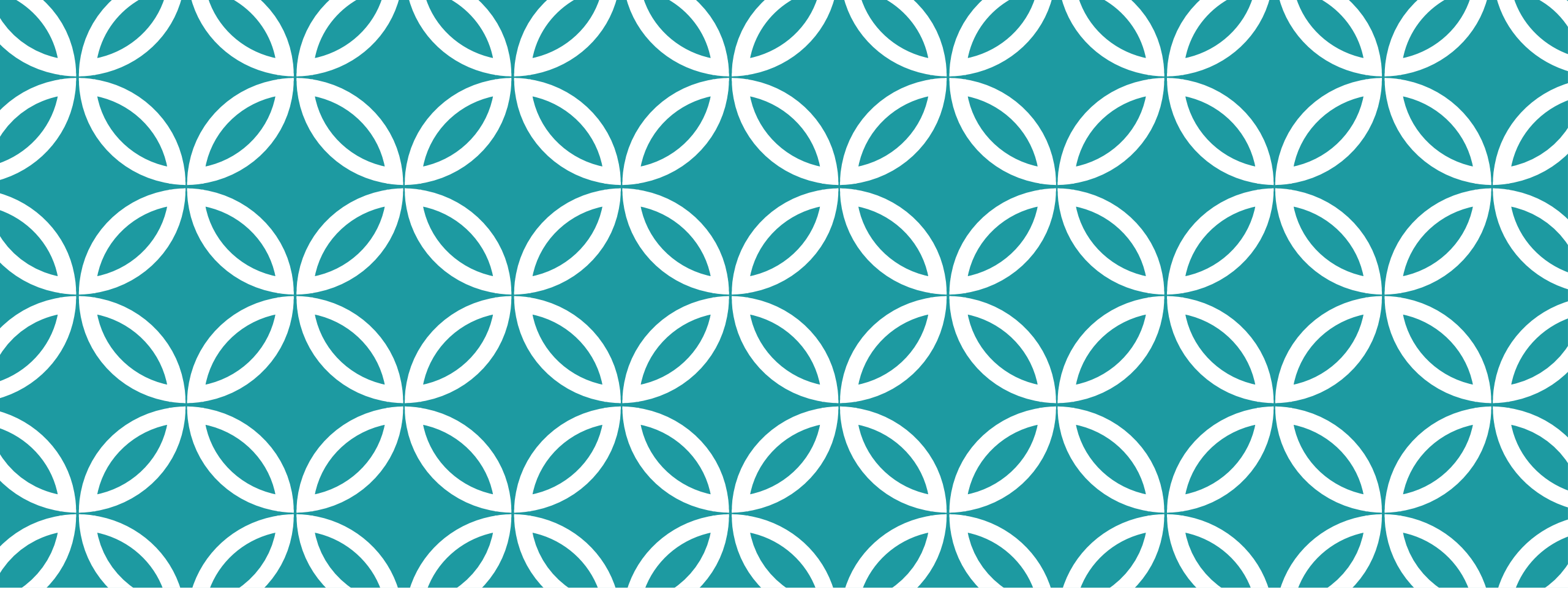




STAT 221- FUNDAMENTALS OF STATISTICAL DATA ANALYSIS

Week-4: Frequency Distributions, Cross Tabulation,
Data Visualization

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FREQUENCY DISTRIBUTIONS |

PROPORTIONS, PERCENTAGES, AND RATIOS

TABLE 1 *Symbols and Formulas*

SYMBOL	MEANING OF SYMBOL	FORMULA
N	The number of cases in a table	
F	The frequency of cases for a particular attribute	
Cf	The cumulative frequency of cases for a group of attributes	Σf
P	The sample proportion	$P = \frac{f}{N}$
π	The population proportion	$\pi = \frac{f}{N}$
%	Percent	$\% = \frac{f}{N}(100)$
$c\%$	Cumulative percent	$c\% = \frac{cf}{N}(100)$
Ratio	The relative frequency of cases across different populations	$Ratio = \frac{f_1}{f_2}$

Proportion: A way to standardize the frequency of cases so that comparisons can be made across populations.

In this formula, f is equal to the number of males who indicate they are satisfied with their social life and N is equal to the total number of males who answered the question. Therefore:

$$P = \frac{f}{N}$$

$$P = \frac{375 \text{ satisfied}}{500 \text{ total}} = \frac{375}{500} = .750$$

Percentages. Percentages are extremely easy to calculate once you know how to calculate proportions. To determine a percentage, first, calculate a proportion and, second, multiply the proportion by 100. That's it! The formula for a percentage is shown below:

$$\% = \frac{f}{N}(100)$$

Ratio: A way to compare the relative frequency of cases across populations.

$$Ratio = \frac{f_1}{f_2}$$

EXERCISE

Suppose we want to know about students' attitudes toward a ban on tobacco use on campus. We decide to survey **500** students, **355 of whom live on-campus and 145 of whom live off-campus**. Of those who live on-campus, 276 indicate that they approve of a tobacco ban and of those who live off-campus, 92 approve of a tobacco ban.

Use this information to answer the following questions.

1. What proportion of all students approve of a tobacco ban?
2. What percent of on-campus students approve of a tobacco ban?
3. What percent of off-campus students approve of a tobacco ban?
4. What is the ratio of on-campus to off-campus students?

Cross tabulation	Approve tobacco ban	Not approve tobacco ban	Row sum
On Campus	276	?	355
Off campus	92	?	145
Column sum	?	?	500

Cross tabulation	Approve tobacco ban	Not approve tobacco ban	Row sum
On Campus	276	79	355
Off campus	92	53	145
Column sum	368	132	500

Cross tabulation (%) ROW PERCENTAGE	Approve tobacco ban	Not approve tobacco ban	Row sum
On Campus	77.75%	22.25%	100
Off campus	63.44%	36.55%	100
Column sum	73.6%	26.4%	

Interpretation?

Cross tabulation (%) COLUMN PERCENTAGE	Approve tobacco ban	Not approve tobacco ban	Row sum
On Campus	75.00	59.85	71.00
Off campus	25.00	40.15	29.00
Column sum			

Interpretation?

CROSS TABULATION OF /GOOGLE DOCS

PIVOT TABLES

Survey Pivot tables			
COUNTA of Do you own a pet by Your Gender			
Do you own a pet	Female	Male	Grand Total
No	25	7	32
Yes	9	4	13
Grand Total	34	11	45

Do you own a pet?	Female	Male	Overall
No	73.53	63.64	71.11
Yes	26.47	36.36	28.89
Total	100.00	100.00	100.00

Suggested



Rows

Add

Do you own a pet?



Order

Ascending



Sort by

Do you own a ...



☒ Show totals

Columns

Add

Your Gender



Order

Ascending



Sort by

Your Gender



☒ Show totals

Values

Add

Do you own a pet?



Summarize by

COUNTA



Show as

Default



I have read the ab...

Your Gender

Your age in years

Place of birth-state

Place of birth-city/...

What is your Moth...

Number of family ...

Do you own a pet?

What is your religi...

What is your moth...

What is your fathe...

The number of cla...

How many books ...

Do you regularly r...

How do you find f...

There a water lillie...

PERCENTAGE , VALID PERCENTAGE AND CUMULATIVE PERCENTAGE

TABLE 3 *Improving Protecting Environment*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	TOO LITTLE	992	22.0	68.6	68.6
	ABOUT RIGHT	365	8.1	25.2	93.8
	TOO MUCH	89	2.0	6.2	100.0
	Total	1446	32.1	100.0	
Missing	NAP	3026	67.1		
	DK	38	.8		
	Total	3064	67.9		
Total		4510	100.0		

Missing values: Either the question did not apply to them (NAP) or they did not provide an answer to the question (DK)

Value labels Descriptive labels for the attributes of a variable.

Frequency A column in a frequency table that shows the number of times a particular attribute occurs.

Percent A column in a frequency table that standardizes frequencies by expressing them as the number of times an attribute occurs per 100 cases. Based on all cases.

Valid percent A column in a frequency table that standardizes frequencies by expressing them as the number of times an attribute occurs per 100 cases. Based on only those cases that provided data.

Cumulative percent A column in a frequency table that shows the percent of cases above or below a certain point or attribute.

EXERCISE

Suppose you are doing a study of student integration into social life on campus and want to find out what percent of students live on campus. Using a survey research method, you ask students about their sex, class standing, and **what percent of their time they spend on campus**. You then test out your questionnaire on a group of students in a dining hall and find the following:

#	SEX	CLASS STANDIN G	TIME ON CAMPUS
1	Male	Freshman	90
2	Male	Junior	75
3	Female	Junior	60
4	Female	Sophomore	85
5	Female	Freshman	95
6	Male	Senior	50
7	Female	Junior	70
8	Female	Senior	60
9	Male	Sophomore	90
10	Female	Junior	70

Using the data from the test of your questionnaire, construct a frequency table for the variable Sex. Be sure to include **value labels, frequencies, percents, valid percents,**
and cumulative percents.

ORDINAL VARIABLE

TABLE 14 *How Satisfied Are You with Your College Experience?*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	Very satisfied	342	68.4	68.4	68.4
	Somewhat satisfied	116	23.2	23.2	91.6
	Not satisfied	42	8.4	8.4	100.0
	Total	500	100.0	100.0	

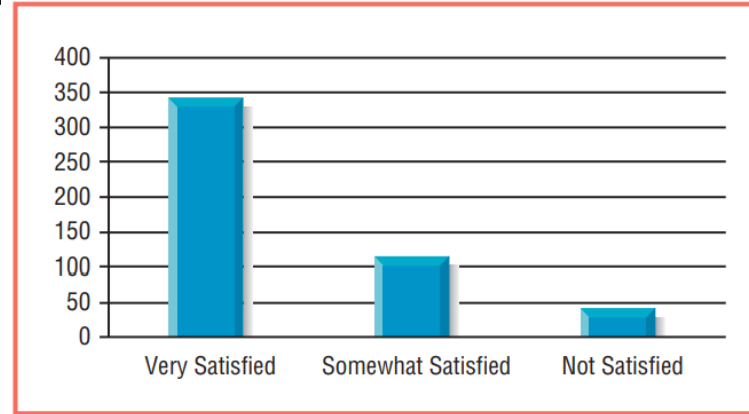
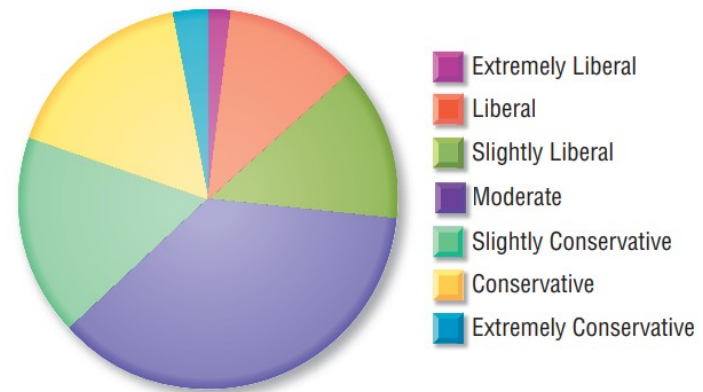
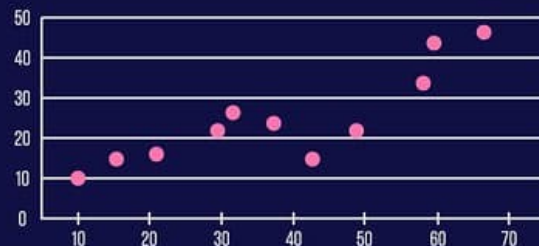
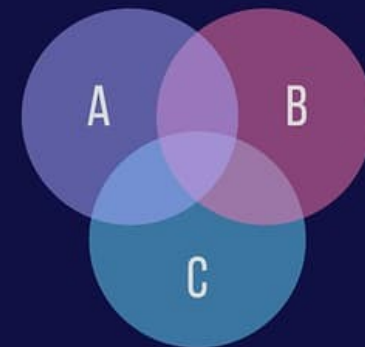


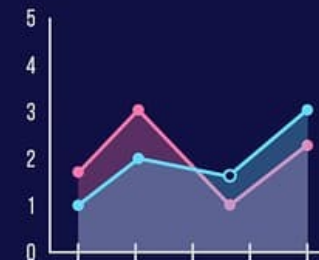
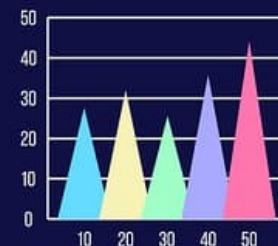
TABLE 16 *Think of Self as Liberal or Conservative*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	Extremely liberal	30	2.0	2.1	2.1
	Liberal	163	10.9	11.3	13.4
	Slightly liberal	193	12.9	13.4	26.7
	Moderate	527	35.1	36.5	63.3
	Slightly conservative	248	16.5	17.2	80.5
	Conservative	241	16.1	16.7	97.2
	Extremely conservative	41	2.7	2.8	100.0
	Total	1443	96.2	100.0	
Missing	DK	48	3.2		
	NA	9	.6		
	Total	57	3.8		
Total		1500	100.0		





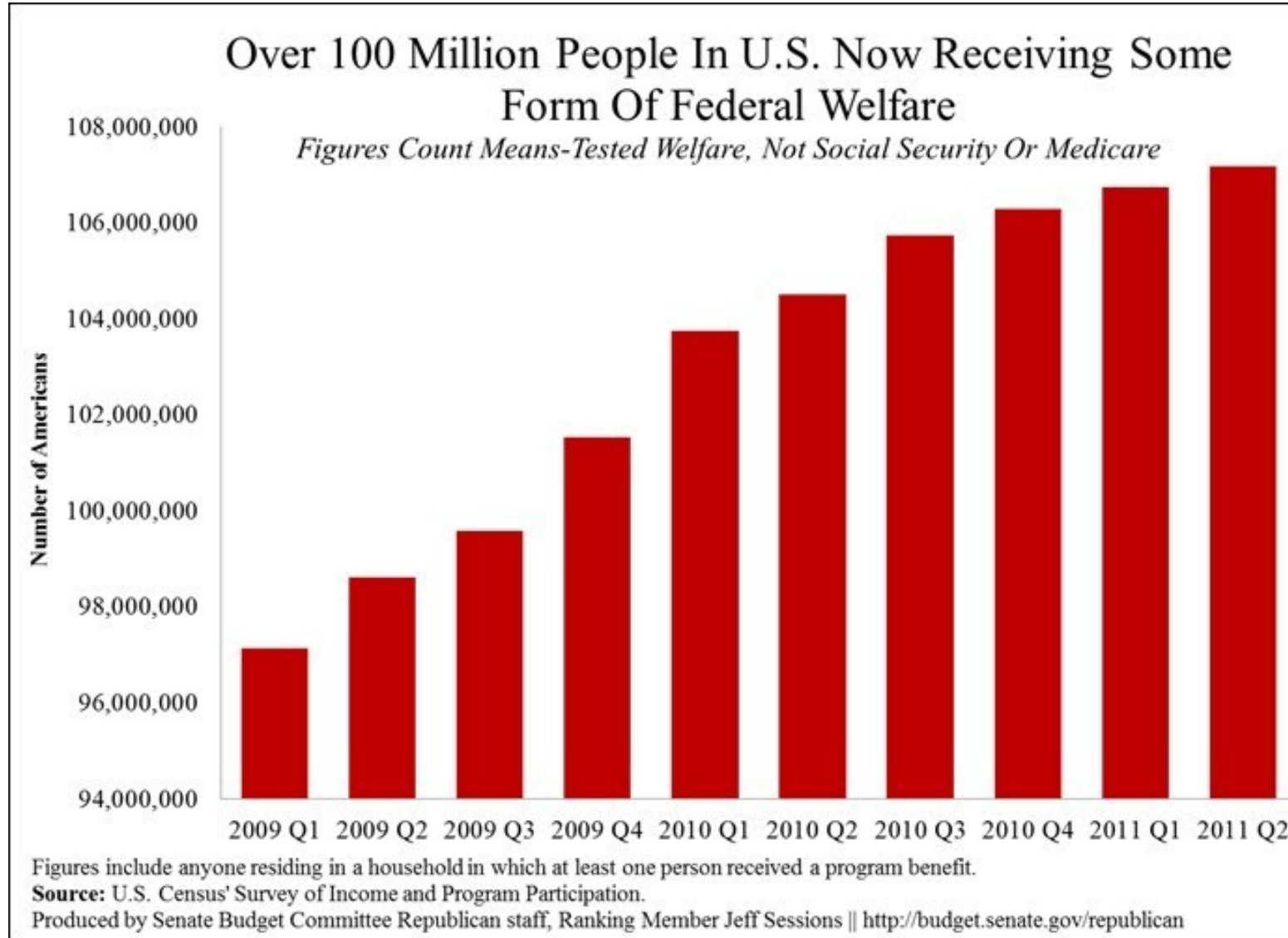
ONE TWO



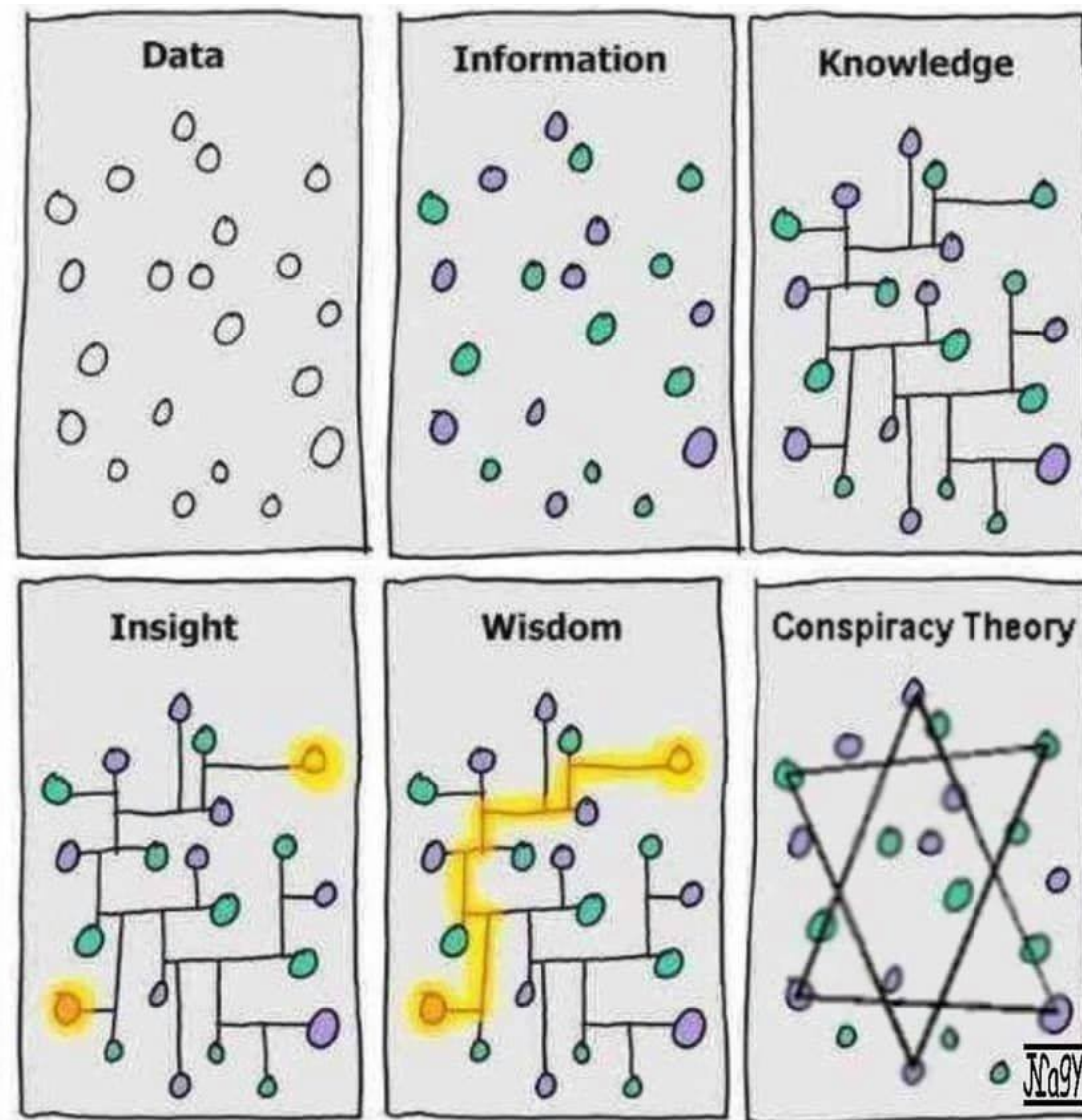
VISUAL REPRESENTATION OF VARIABLES



DATA DECEPTION



DIFFERENTIATING : DATA, INFORMATION, KNOWLEDGE, INSIGHT AND WISDOM, AND CONSPIRACY



DATA VISUALIZATION

Data visualization is a graphic representation that expresses the significance of data. It reveals **insights and patterns that are not immediately visible in the raw data**. It is an art through which information, numbers, and measurements can be made more understandable. According to (Friedman [2008](#)):

The primary goal of data visualization is to communicate information clearly and effectively through graphical means. It does not mean that data visualization needs to look boring to be functional or extremely sophisticated to look beautiful. To convey ideas effectively, both aesthetic form and functionality need to go hand in hand, providing insights into a rather sparse and complex data set by communicating its key-aspects more intuitively.

https://mschermann.github.io/data_viz_reader/introduction.html#what-is-data-visualization

WHY GOOD DATA VISUALIZATION IS SUCH A POWERFUL TOOL?

Intuitive: Presenting a graph as a node-link structure instantly makes sense, even to people who have never worked with graphs before.

Fast: It is fast because our brains are great at identifying patterns, but only when data is presented in a tangible format. Armed with visualization, we can spot trends and outliers very effectively.

Flexible: The world is densely connected, so as long as there is an interesting relationship in your data somewhere, you will find value in graph visualization.

Insightful: Exploring graph data interactively allows users to gain more in-depth knowledge, understand the context and ask more questions, compared to static visualization or raw data

BASIC CRITERIA THAT A DATA VISUALIZATION SHOULD SATISFY TO BE EFFECTIVE

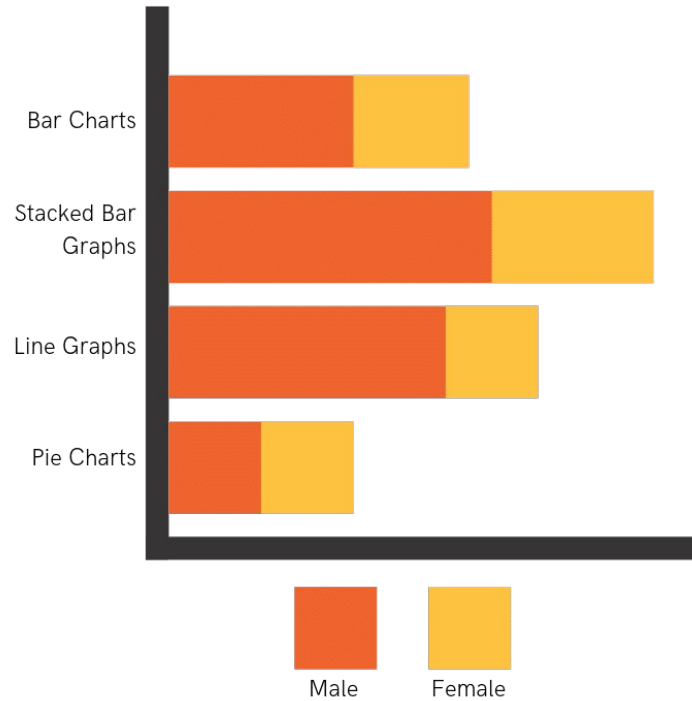
Criteria	Description
Informative	The visualization should be able to convey the desired information from the data to the reader.
Efficient	The visualization should not be ambiguous.
Appealing	The visualization should be captivating and visually pleasing.
Interactive and Predictive (Optional)	The visualizations can contain variables and filters with which the users may interact to predict results of different scenarios.

Pittenturf, Chris. 2018. "What Is Data Visualization and Why Is It Important?" <https://data-visualization.cioreview.com/cxoinsight/what-is-data-visualization-and-why-is-it-important-nid-11806-cid-163.html>.

NOMINAL VAI

Part of a whole

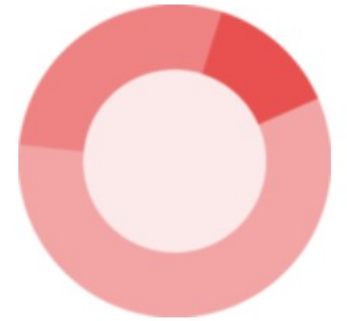
Favorite Chart Types



Grouped and Stacked
barplot



Treemap



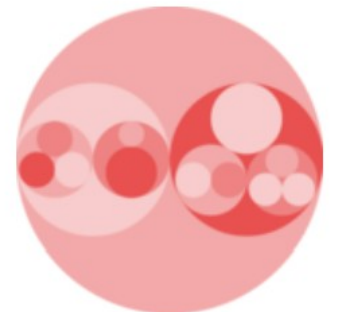
Doughnut



Pie chart



Dendrogram



Circular packing

RANKING (BOTH NOMINAL AND ORDINAL)

Ranking



Barplot



Spider / Radar



Lollipop



Circular Barplot

INTERVAL/RATIO- UNIVARIATE

Distribution



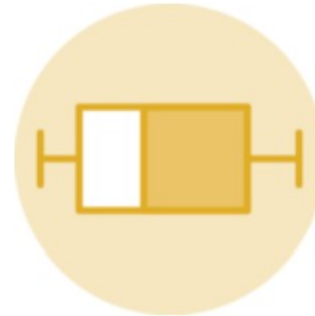
Violin



Density



Histogram



Boxplot

RATIO- BIVARIATE

Correlation



Scatter



Heatmap



Correlogram



Bubble



Connected scatter



Density 2d